

Explaining Kafka



Introduction

Let's imagine you are at a party where people want to share exciting **news or stories** with each other.

Normally, you would go around talking to each person individually, which can be time-consuming.



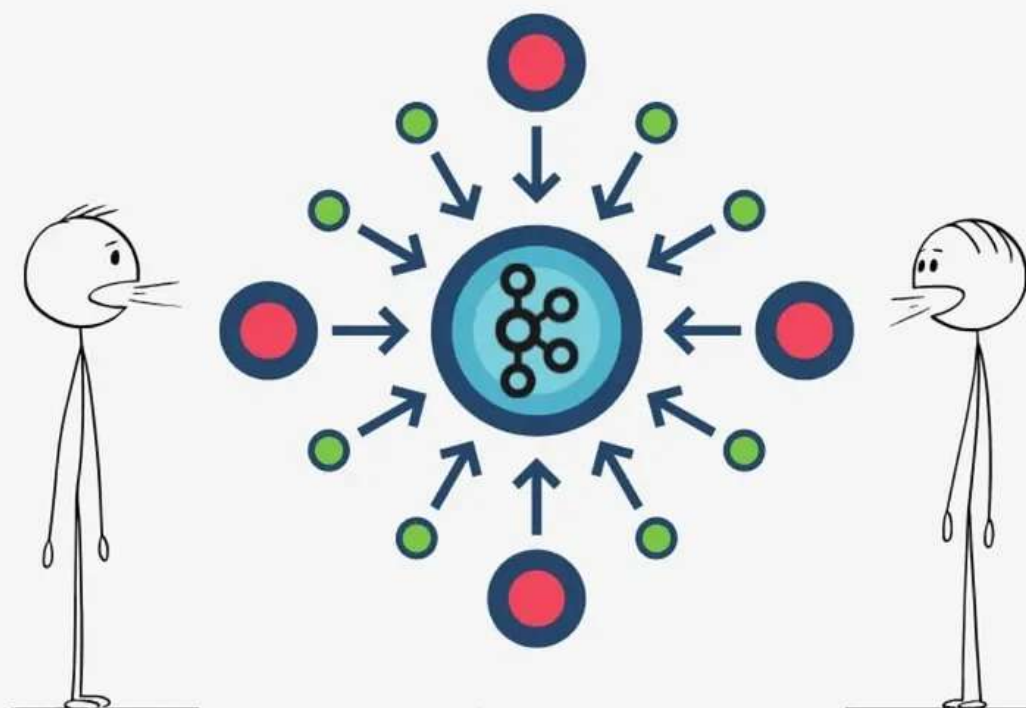
But here comes **Apache Kafka**, the party organizer!



Kafka: Centralizing the Buzz

Apache Kafka is like a central communication hub at the party.

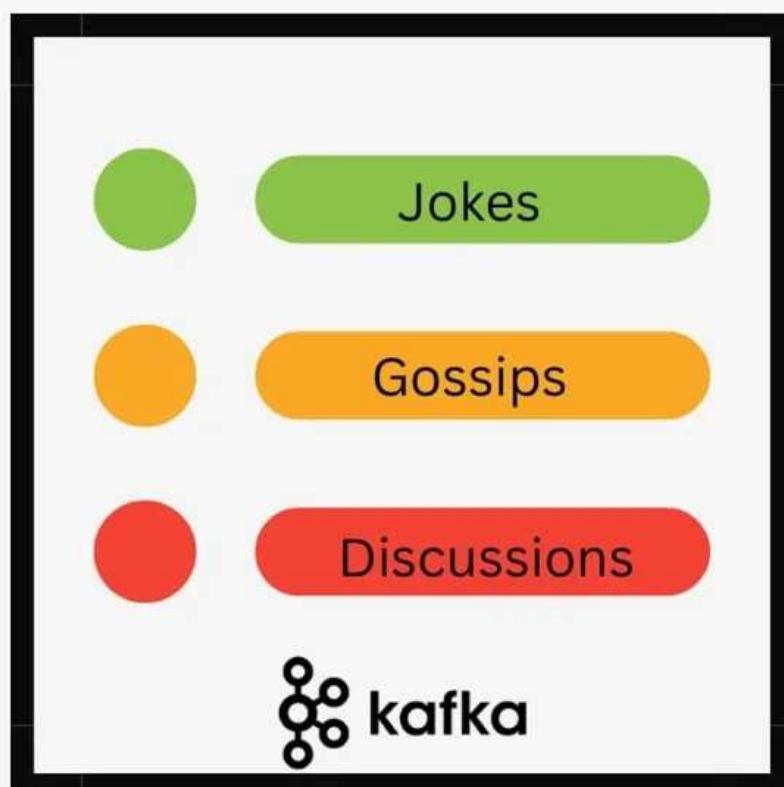
Instead of everyone talking to each other individually, they can now come to the hub and share their news or stories.



Kafka Topics

The hub, which is Apache Kafka, **collects all the messages** and organizes them into **different categories or topics**.

It keeps track of **who** said **what** and **when**.



Topic-Based Message Distribution

Now, imagine you have a **group** of friends who are really interested in **funny jokes**. Apache Kafka makes it easy for you to share all the funny jokes with this specific group.

Kafka will tag those with the **topic "Funny Jokes."** Your friends can then come to the hub and read all the funny jokes people shared with each others.



What is Apache Kafka?

Similarly, in the world of software development, **Apache Kafka** is a **distributed streaming platform** that acts as a **central hub** for handling **large amounts of data**.

It allows different systems and applications to communicate and **share data** with each other in **real-time**.

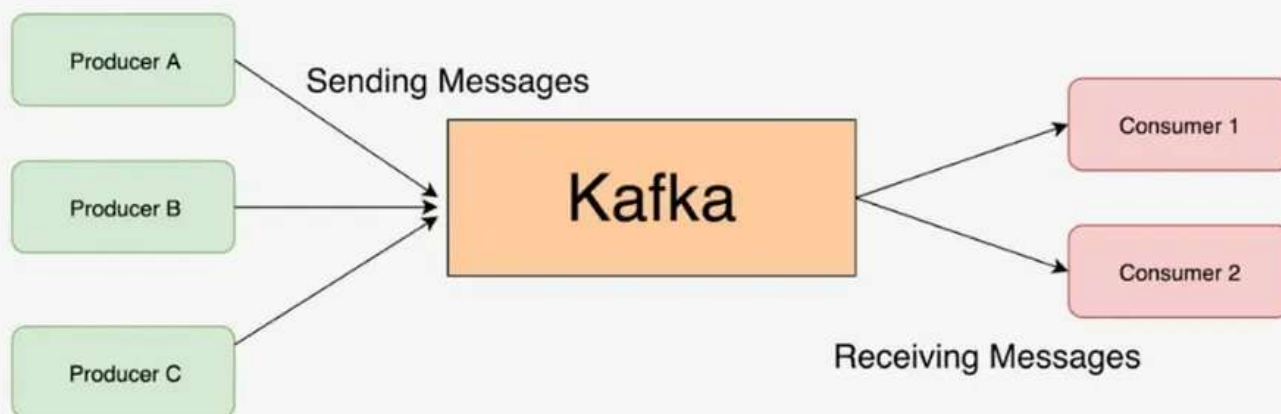


Producers and consumers

Data in Kafka is organized and stored into **topics**.

In Kafka terminology, those **that write data** to topics are called **producers**, while those **who read data** from them are called **consumers**.

Having data categorized in topics helps producers and consumers to exchange only data of interest.



Kafka advantages

Apache Kafka is known for its:

high scalability



You can easily
increase number
of computers

fault-tolerance and low latency.



No data is lost
even when
accident happens



It's very fast!



Which **companies** use **Kafka**?

Kafka is widely used by companies like **LinkedIn**, **Netflix**, and **Uber** for real-time data streaming, log aggregation, and event-driven architectures. 🚀

